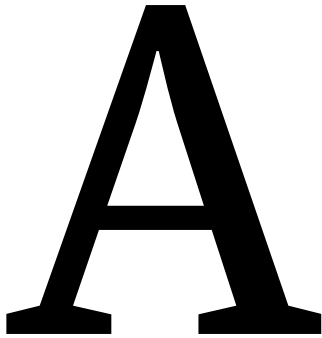




Artificial intelligence is one of the biggest disruptions the world is grappling with right now. From businesses to jobs to social behaviour, AI's impact can be felt everywhere. At the *Business Today India@100 Summit* held recently, India Today Group Vice Chairperson *Kalli Purie* caught up with AI expert Soumitra Dutta, who has a PhD in the technology and has been tracking it for three decades, to discuss a range of issues that this disruption gives rise to. In a session called 'Managing with AI', Dutta, the Peter Moores Dean and Professor of Management at Saïd Business School, University of Oxford, spoke extensively about how AI will affect jobs and skill sets, the need for regulations, and the concept of responsible AI. Edited excerpts:

Q: You've talked about the Moore's Law of computer processing power and how it doubles every 18 months and that there's a tipping point for technology. You have a very good visual related to Lake Michigan to explain this. Please explain what that means and where you think we are right now.

A: We all know that we are living in exponential times, and often that is substantiated by saying that technology is increasing exponentially, both in terms of computing power, and in terms of data and other similar aspects... But what exactly does it mean in terms of trends over time? So this is a small visual... [that] shows the filling up of a lake in the US, Lake Michigan, whose volume is roughly the capacity of

the human brain in computations per second. And it fills in water at Moore's Law frequencies. So you begin with one fluid ounce; in 18 months, it doubles; in another 18 months it doubles and so on. On the top right, [there is] a time scale... [that] goes from 1940 to 2025... so the law is carrying on doubling for 70 years. And then, in 2010, suddenly it explodes. And that's where the hockey stick part of the exponential curve kicks in... Moore's Law [for] modern microprocessors started sometime in the 1960s. So in 2023, we are roughly 63 years into the modern microprocessor age... which means what we have seen in the last 5-10 years is nothing compared to what we will see in the next 5-10 years... it's going to explode.

And one reason why we are seeing all these great applications of AI right now is because of the additional computational power, additional data that's available right now. So what will happen in the next 5-10 years, it really remains for us to discover and, at the same time, be amazed by.

Q: So just taking on from there, humans are more linear, they think linearly, while technology is moving exponentially. So we're on two different curves, right? How do we even compete with technology when it starts moving at this rate?

A: I don't know whether you can compete with technology, per se, but what you can try and do is help to create better lives for people, help to create better organisations and help to become more competitive in whatever we are trying to achieve in the organisation.

I think the challenge of any kind of digital/AI transformation... is essentially combining two things. The first part is building forwards. So all of us have a business... an organisation... [and] you have to apply technology to improve what you're doing things today.

What is much harder, at the same time much more exciting, is creating the future. Now, what will happen in five years' time, 10 years' time? We don't know... what you can do is try to experiment with ideas, with concepts of business models and different kinds of situations, and learn from it and work backwards. So you have to do experiments and then try to design your organisation so that if it shows signs of success, you move towards that in some reasonable path. So this duality, the dual challenge of building forwards and creating backwards is the challenge out here.

But creating backwards is very hard. Because sometimes it means you have to come up with experiments that are disruptive, that go against the current